

ZEXUAN WANG

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Education

University Of California, Riverside

Master of Science in Computer Engineering

Beginning Sep. 2026

Riverside, CA

University Of California, Riverside

Bachelor of Science in Computer Engineering, GPA: 3.75

Sep. 2022 – Jun. 2026

Riverside, CA

Experience

UCR Embedded Systems Research Lab

Undergraduate Research Assistant

December 2024 – Present

University Of California Riverside

- Research TCAM-integrated memory systems for accelerating tokenization workloads in AI applications.
- Integrate and evaluate custom memory architectures within the gem5 simulator across ARM and RISC-V platforms.
- Profile memory access behavior, bandwidth, and instruction mix via gem5 ROI and m5 instrumentation.
- Develop cross-language integrations (Rust and C/C++) to support custom hardware models within real-world software stacks.

UCR Residential Education

Residential Advisor

September 2023 – Present

University Of California Riverside

- Maintain accurate resident and facility records while supporting administrative operations.
- Plan and facilitate community-building and academic support events for residential students.
- Provide conflict resolution and crisis support for residents, coordinating with professional staff when necessary.

StoneRose Management

Account Manager

June 2023 – September 2023

Diamond Bar, CA

- Developed and executed data-driven strategies with AT&T to align with client objectives
- Analyzed account performance metrics and presented recommendations to support client retention and growth.
- Maintained client relationships through proactive communication and issue resolution.

Projects

TCAM-Accelerated Tiktoken (Research Project) | C++, C, Rust, Python, VS Code

- Implemented a TCAM-backed data structure in Rust Tiktoken to accelerate lookup operations in BPE tokenization.
- Performed C-injection to bridge Rust Tiktoken code with C++ TCAM models, resolving complex type and ABI constraints.
- Migrated Tiktoken simulations from ARM to RISC-V in gem5, reducing simulation time by 5x while preserving correctness.
- Analyzed DRAM and instruction-level statistics to identify memory bottlenecks and assess TCAM performance potential.

GamePod | C++, C, Micro Controllers

- Built an open-source retro handheld gaming console on the ESP32 with single-player, multiplayer, and Game Boy ROM emulation.
- Developed graphics drivers and rendering pipelines for a 320x240 SPI LCD display, implementing sprite rendering and game object visualization
- Implemented multiplayer Tron game logic: player movement, collision detection, and elimination mechanics.
- [Research Poster](#)

Dance Dance Karaoke | Python, PoseNet, Jetson Nano, Pygame, Google Cloud

- Developed an open-source karaoke and dance platform on the Jetson Nano, combining real-time choreography analysis, lyric analysis, pitch analysis, and cloud-hosted data management
- Integrated PoseNet for real-time body keypoint detection and skeletal tracking during gameplay.
- Designed an angle-based pose comparison algorithm that evaluated player movements against reference choreography sequences to generate dance accuracy scores.
- [Research Poster](#)

Technical Skills

Languages: C++, C, Python, Rust, Java, Assembly, GDScript

Developer Tools: Git, GitHub, VS Code, PlatformIO

Technologies: ESP32, Jetson Nano, AVR Microcontrollers, gem5, Godot, Linux